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SOFTWARE **ENGINEERING**

M03 — Use Cases, System Boundary & Activity Modelling

Sessions 05 + 06 · one M03 deck

Table of Contents — M03

Sessions 05 + 06

01 Project position

02 Actors & stakeholders

03 System boundary

04 Use cases

05 Task 2

06 Textual use case

07 Activity diagrams

08 Consistency & traceability

09 Tasks 3–4

10 Export & close

Submission	Required coverage	Points
Checkpoint 1	M01–M02	20
Checkpoint 2	M01–M04	20
Checkpoint 3	M01–M06	20
Checkpoint 4	M01–M08	20
Checkpoint 5 — final project	M01–M10	20
Total	**10 modules**	**100**

MOODLE CHECKPOINTS**

Checkpoint 1
M01–M02

Checkpoint 2
M01–M04

Checkpoint 3
M01–M06

Checkpoint 4
M01–M08

Checkpoint 5
final project*

*final project submission - 1 week after our last class

*Always export your project before leaving the lab. A Moodle checkpoint is not the same thing as a progress backup.

Syllabus topic	Coverage
Project vision, lifecycle and feasibility	M01
Business, functional and non-functional requirements	M02
System functions and information flows	M03–M04
Reliability, availability, security and fault tolerance	M05
Physical infrastructure and deployment	M06
UML diagrams and design patterns	M03, M07–M08
CASE tools	Website modelling tools used in required diagram tasks
Project management and professional practice	M09
Testing methods and practical exercises	M10: test plan, six test cases and two manual walkthroughs

QuickBite project - criteria / requirements

1. **Work in teams of 3**

Submit one agreed project ZIP per team. Everyone should contribute and be able to explain the team's decisions.

2. **Each module is worth 10 points**

Each module has 5 criteria worth 0–2 points each, published with its instructions:

- 2 points: complete, clear and justified.
- 1 point: partly correct, but incomplete or unclear.
- 0 points: missing or fundamentally incorrect.

3. **Different answers can earn full marks**

We assess your reasoning, use of the project brief and consistency between your answers. There is no single ideal solution.

4. **Focus on required tasks**

Optional tasks are extra practice. Skipping them causes no penalty; completing them adds no points.

5. **Submit your work through Moodle**

Save your answers on the website, export the latest project ZIP and upload it by the assignment deadline. "Saved" on the website confirms storage - it is not a grade.

6. **Project total: 100 points**

M01 is worth 10 points across Sessions 1 and 2 together. The project contributes to the laboratory component;

M

M03 → S05 + S06

Task 2 → choose one use case from your diagram
Task 3 → describe its normal flow + alternative/failure flow
Task 4 → model the same flow as an activity diagram

Overview → detailed description → visual workflow

Where are we in the project?

M03 models behaviour already justified by earlier work.

M01

Problem · vision
stakeholders · scope
business objectives

→

M02

FR / NFR · constraints
acceptance criteria
reviewed baseline

→

M03

Actors · use cases
textual use case
activity flow

→

M04

Context diagram · DFD
information structures

M03 should explain behaviour — not invent a new product.

Recap from M02

What should already exist before you start M03?

REQUIREMENT IDs

Stable identifiers for your functional and quality requirements.

Later models should reference the same IDs.

USER-ORIENTED SCENARIO

One important interaction with a normal outcome and a failure OR boundary outcome.

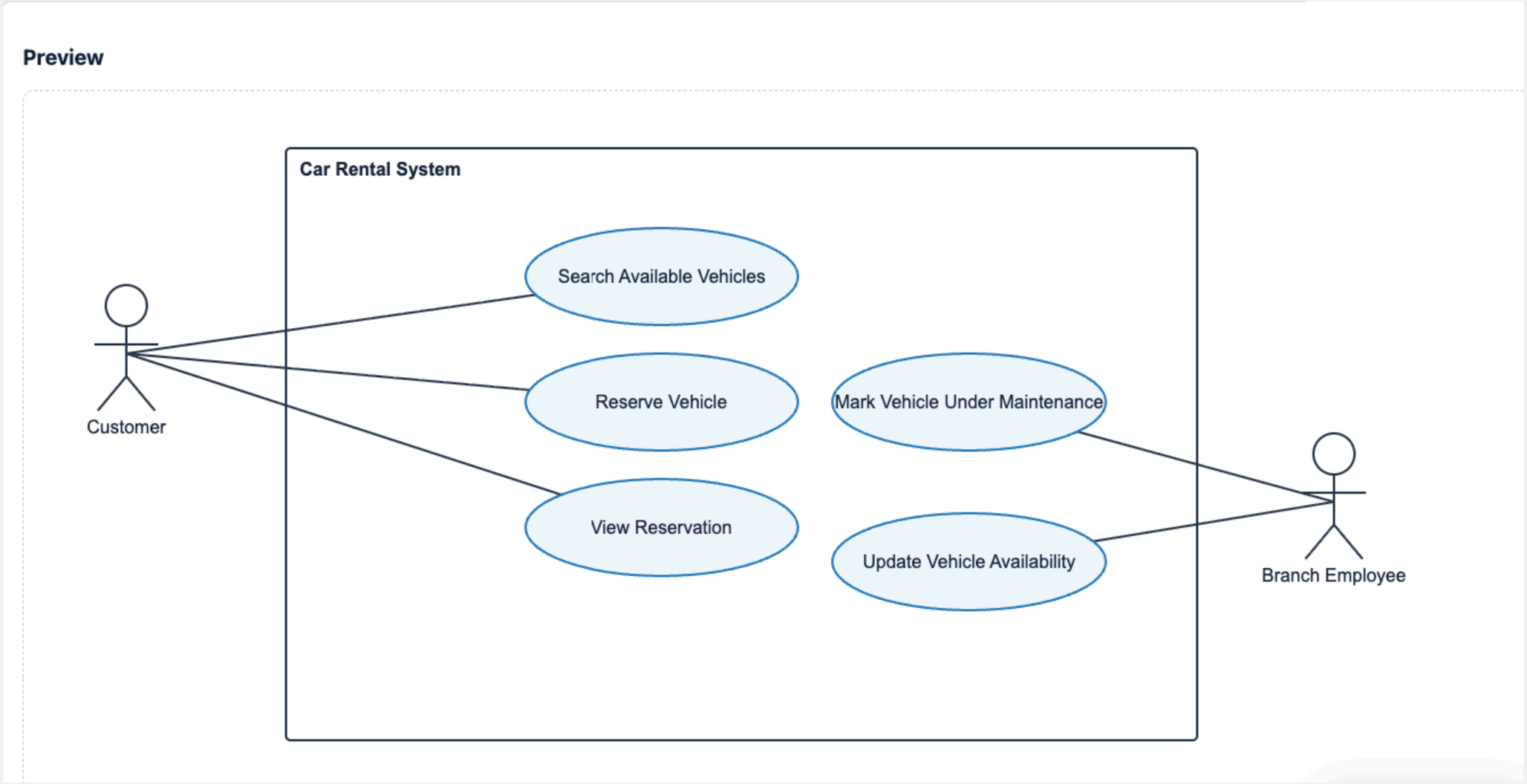
This becomes useful input for choosing a use case.

REVIEWED BASELINE

Both M01.5 objectives checked for coverage. Requirements reviewed for clarity and testability.

Today: turn selected behaviour into models that remain consistent with this baseline.

What is a use case?



Use-case diagram shows actors, not all stakeholders

What is a use case?

A use case represents an actor goal achieved through interaction with the system.

GOOD

Search Available Vehicles

Reserve Vehicle

Mark Vehicle Under Maintenance

View Reservation

Update Vehicle Availability

WEAK

Search Screen

Reserve Button

Database Check

Process Data

Handle Reservation

RULE OF THUMB

Use a verb + object that expresses an actor goal or outcome.

Describe the actor goal — not the interface element or implementation step.

A use case should describe what the actor wants to achieve — not which screen they open or which internal operation the software performs.

Use cases answer: “What goal does this actor achieve through the system?”

Actor ≠ stakeholder

The distinction depends on direct interaction with the system boundary.

STAKEHOLDER

Anyone affected by, interested in, funding, regulating or depending on the system.

A stakeholder may never directly use the system.

ACTOR

A role, person, organisation or external system that interacts directly with the system being modelled.

Actors stay outside the system boundary.

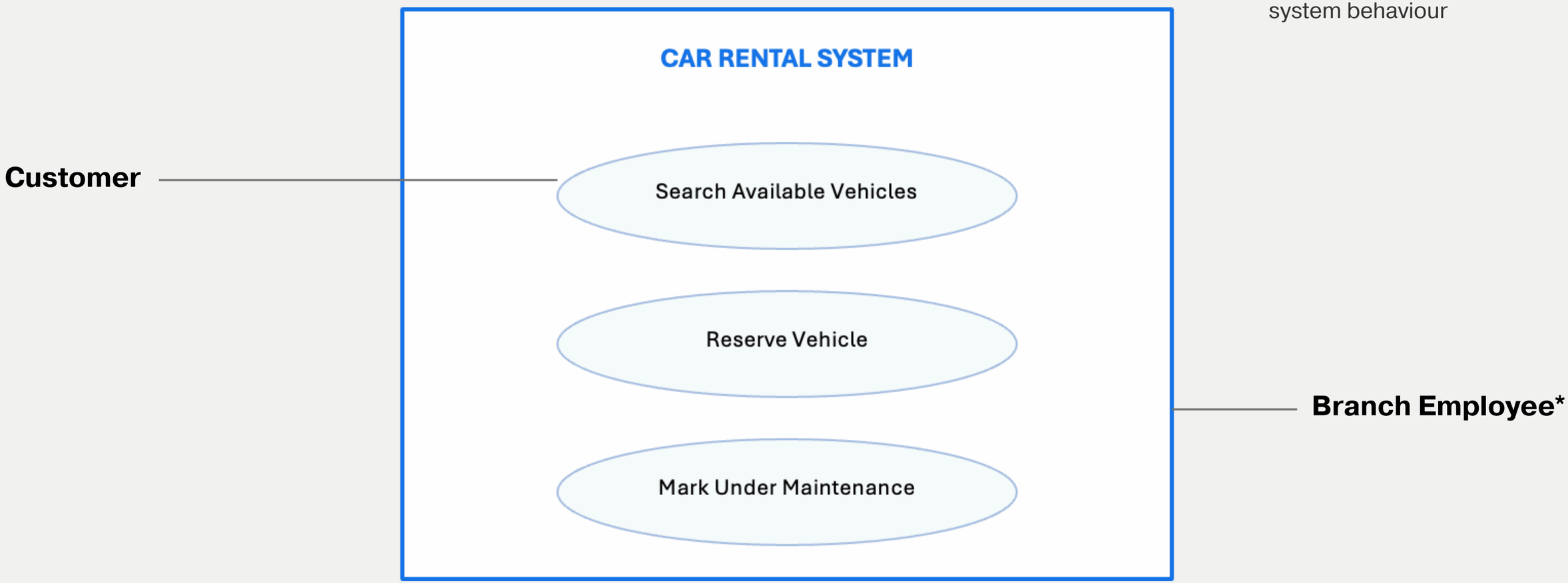
Stakeholder	Need	Direct system interaction?	Actor?
Customer	Find and reserve a vehicle	Yes	✅
Branch Employee	Maintain accurate vehicle availability	Yes	✅
Branch Manager	Improve vehicle utilisation	No	❌

Car Rental examples Customer → actor • Branch Employee → actor • Branch Manager → actor **only** if they interact with the system

System boundary

What belongs inside the system — and what remains outside?

*we don’t necessarily place stakeholders that are not actors (without linkages to use cases) on UML diagrams unless it’s important/crucial for the demonstration of the system behaviour



Boundary = a responsibility decision

Use-case diagram — core notation

Keep the notation simple enough to communicate goals and boundaries.

ACTOR

External role or external system.

Placed outside the system boundary.

USE CASE

Goal-level behaviour.

Shown as an ellipse inside the boundary.

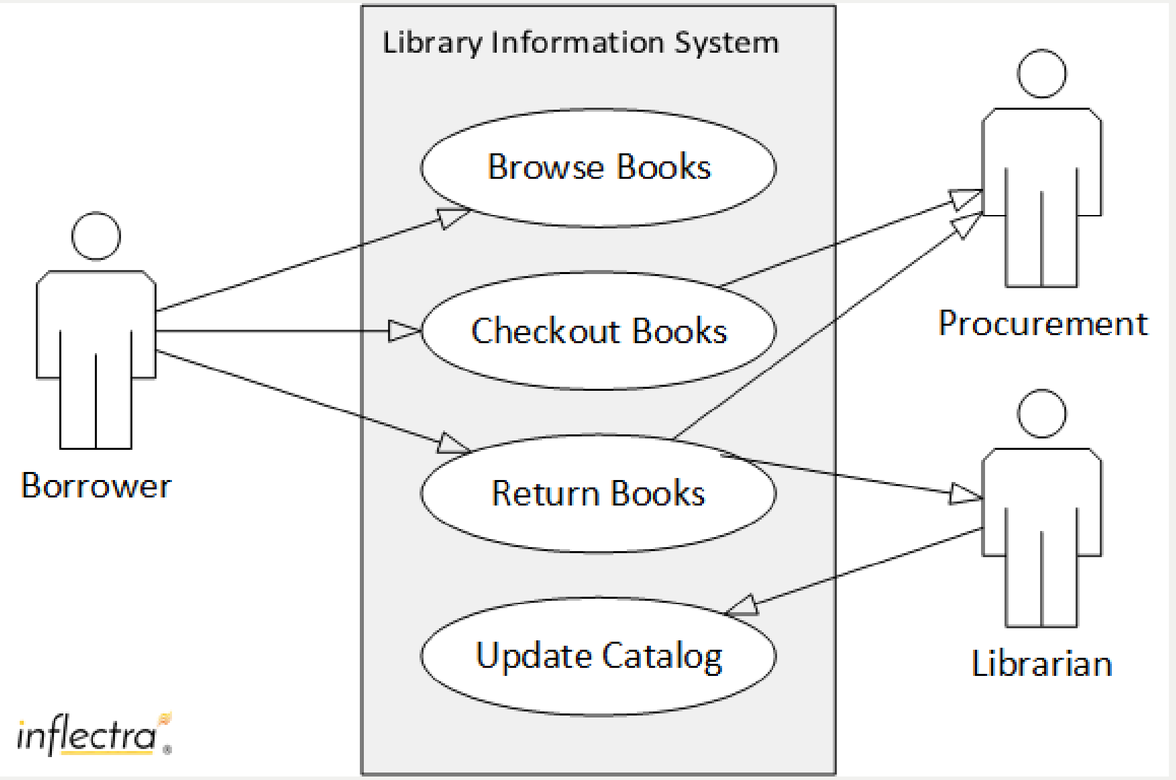
ASSOCIATION

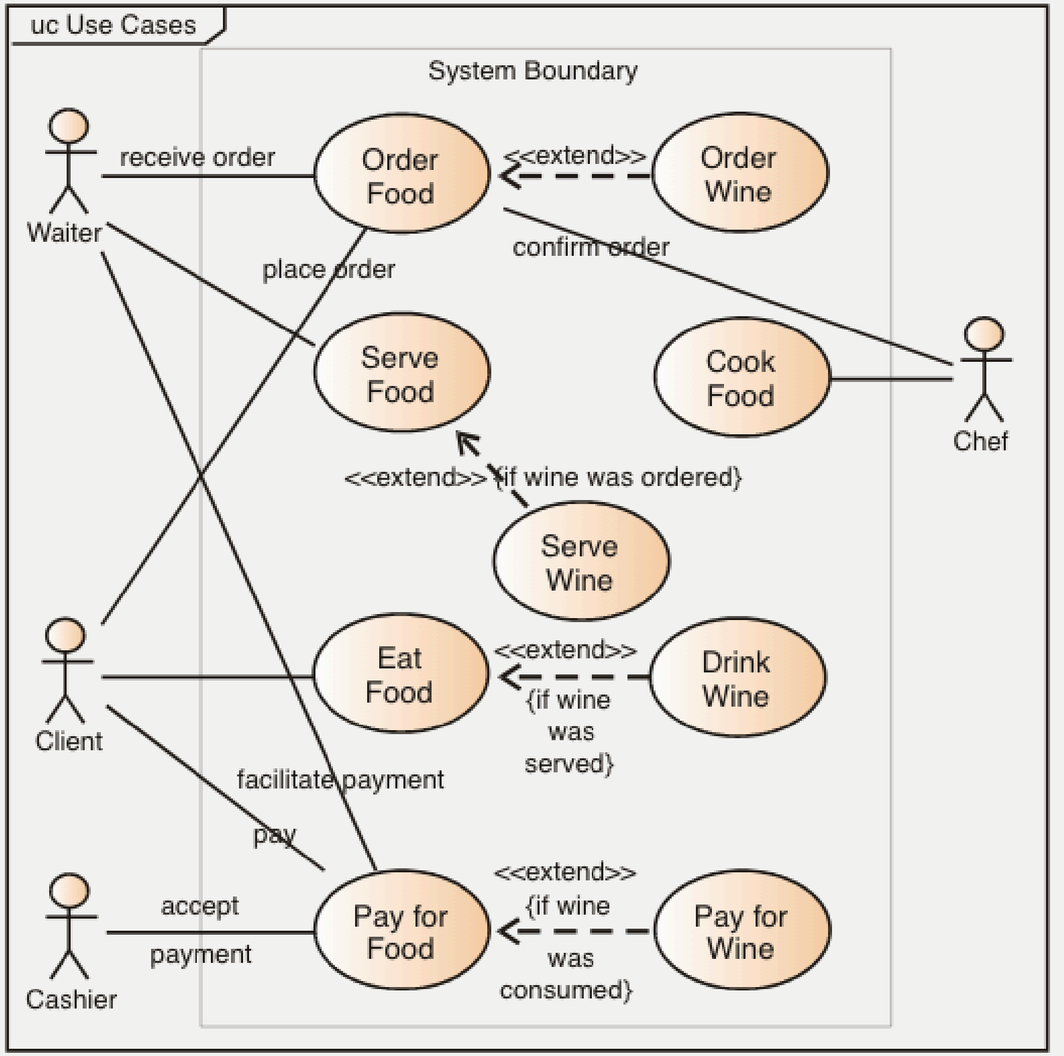
Line showing that an actor participates in a use case.

BOUNDARY

Rectangle naming the system and separating inside from outside.

«include» / «extend» are not required for the current M03 task. Use them only if you can justify them.


Library Information System
Browse Books
Checkout Books
Return Books
Update Catalog
Borrower
Procurement
Librarian
inflectra


uc Use Cases
System Boundary
Waiter
receive order
Order Food
<<extend>>
Order Wine
confirm order
place order
Serve Food
Cook Food
Chef
<<extend>>
{if wine was ordered}
Serve Wine
Client
Eat Food
<<extend>>
{if wine was served}
Drink Wine
facilitate payment
pay
Cashier
accept payment
Pay for Food
<<extend>>
{if wine was consumed}
Pay for Wine

A diagram is not enough

How to write a use case?

1. **Describe your system**
2. **Identify the actors**
3. **Define your actor’s goals**
4. **Create a scenario**
5. **Consider alternate flows**
6. **Repeat steps 2-5 to compile your use case**

From requirement to use case

Traceability helps explain why the model exists.

FR-01

The system shall search available vehicles for a selected branch and rental period.

→

ACTOR

Customer

→

USE CASE

Search Available Vehicles

Use cases are primarily derived from functional requirements

Functional requirements help identify use cases.
Non-functional requirements constrain how those use cases should be delivered.

FR → actor goal → use case
NFR → quality constraint on that use case

Traceability is not necessarily one-to-one: one use case may support several requirements.

QuickBite — Required Task 2

Create one use-case diagram for your agreed scope.

DELIVERABLE

Create one use-case diagram for your agreed scope.

WHERE TO SAVE

Write on this task card and click Save answer.

Models / CASE → Save name → M03_usecase

Rationale: M03 Report → Actors & Use-case Rationale.

MINIMUM REQUIREMENTS

- Show a system boundary
- Include relevant external actors
- Include 4–6 goal-level use cases
- Save the model
- Briefly explain your actor choices

COMPLETE WHEN Your saved work meets the checklist above. Check it yourself; the website does not grade correctness.

A diagram is not enough

The use-case diagram gives the overview; the textual specification carries behavioural detail.

UC ID
Stable identifier, e.g. UC01

Primary actor
Who initiates or primarily pursues the goal?

Goal
What outcome does the actor want?

Trigger
What event starts the interaction?

Precondition
What must already be true?

Main flow
Numbered normal sequence

Alternative / failure
Meaningful non-normal path

Postcondition
What is true after the interaction ends?

Requirement IDs
Which baseline requirements justify the behaviour?

Trigger vs precondition vs postcondition

These three are often confused.

TRIGGER
The event that starts the interaction.

Car Rental: customer starts a search for a branch and rental period.

PRECONDITION
A state that must already be true before the use case begins.

Example: the rental period is valid / required search inputs are available.

POSTCONDITION
A state that is true after the use case finishes.

Example: matching vehicles are shown, or no availability is reported.

Do not write the trigger as if it was a precondition.

Car Rental — textual use-case example

UC01 · Search Available Vehicles

CONTEXT

Primary actor: Customer

Related: FR-01; QPR-01 where relevant

Goal: find vehicles available for the selected branch and rental period

Trigger: customer requests a search

Precondition: The system is available and the customer can access the search function.

MAIN FLOW

1. Customer provides branch and rental period.
2. System checks vehicle availability.
3. System returns matching available vehicles.
4. Customer can review the result.

ALTERNATIVE

No vehicles match the selected branch and period.

→ System reports that no vehicle is available.

Postcondition: a search result is displayed.

The textual use case describes interaction — it is not a UI click-by-click script.

QuickBite — Required Task 3

Describe one use case from your diagram.

DELIVERABLE

Describe one use case from your diagram.

WHERE TO SAVE

Write on this task card and click Save answer. It is copied into the module report and included in your project ZIP.

MINIMUM REQUIREMENTS

- Include actor, goal, precondition, normal steps and outcome
- Add one alternative or failure flow
- Link to a requirement

COMPLETE WHEN Your saved work meets the checklist above. Check it yourself; the website does not grade correctness.

Activity diagram — what does it show?

An activity diagram models workflow and control flow. It should match the textual use case.

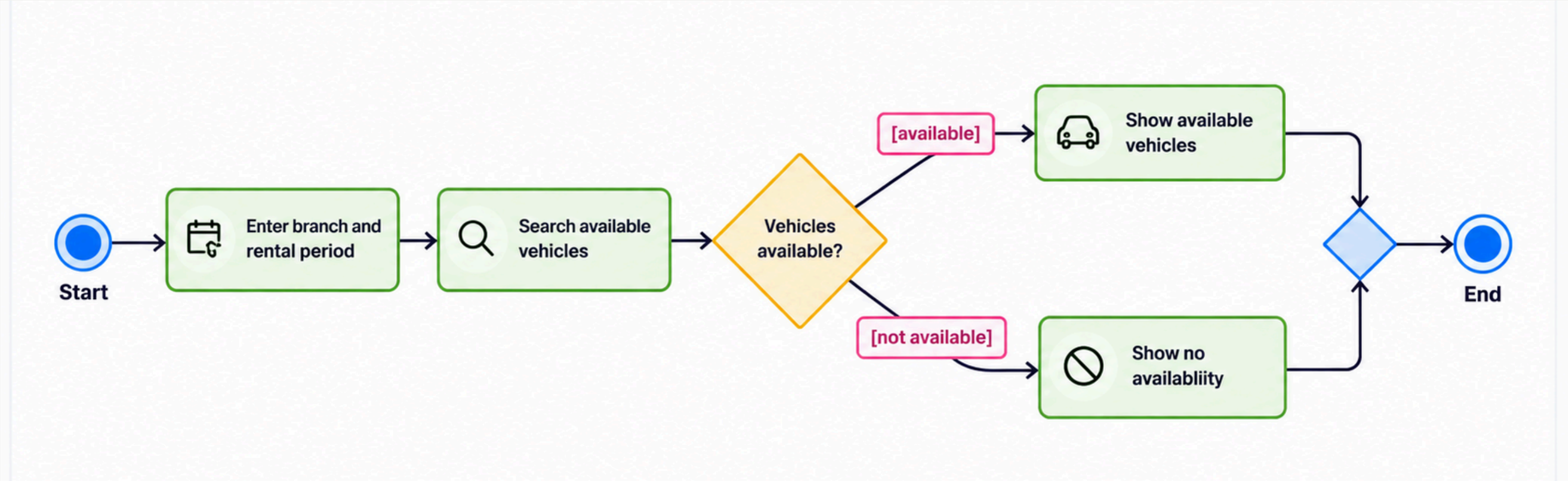
START / END
Where the modelled workflow begins and finishes.

ACTION
A step or piece of behaviour.

DECISION
A point where the flow branches according to a condition.

GUARD
Condition written on an outgoing path, e.g. [available].

MERGE
Brings alternative paths back together when needed.



Activity diagrams show behaviour flow — not stored data structures.

Decisions and guards

A decision is useful only when the outgoing conditions are clear.

WEAK

Availability?

→ Yes
→ No

The labels depend on context and are easy to misunderstand when the model grows.

BETTER

Check availability

→ [vehicles available]
→ [no vehicles available]

The guards state the conditions directly.

Guards should be understandable and testable — and should cover the alternatives you intend to model.

Consistency across M03 artifacts

Different models should tell the same story at different levels of detail.

REQUIREMENT ID

Why does this behaviour exist?

→

USE CASE

Who wants what goal?

→

TEXTUAL FLOW

How does the interaction unfold?

→

ACTIVITY PATH

How does control branch and proceed?

If one artifact adds a new rule, either justify it from the baseline or record it as an unresolved question.

QuickBite — Required Task 4

Create one activity diagram for the same use case.

DELIVERABLE

Create one activity diagram for the same use case.

WHERE TO SAVE

Write on this task card and click Save answer.

Models / CASE → Save name → M03_activity

Rationale: M03 Report → Activity-model Rationale.

MINIMUM REQUIREMENTS

- Show start/end and actions
- Include at least one decision with labelled guards
- Check that normal and alternative paths match your textual use case

COMPLETE WHEN Your saved work meets the checklist above. Check it yourself; the website does not grade correctness.

Common mistakes

Use this as a final self-check before exporting.

ACTOR?

Internal component shown as an actor.

GOAL?

Screen/button used as a use-case name.

BOUNDARY?

System responsibilities are unclear.

FLOW?

Textual and activity paths contradict each other.

GUARD?

Decision paths have vague or missing guards.

TRACE?

Modelled behaviour has no requirement support.

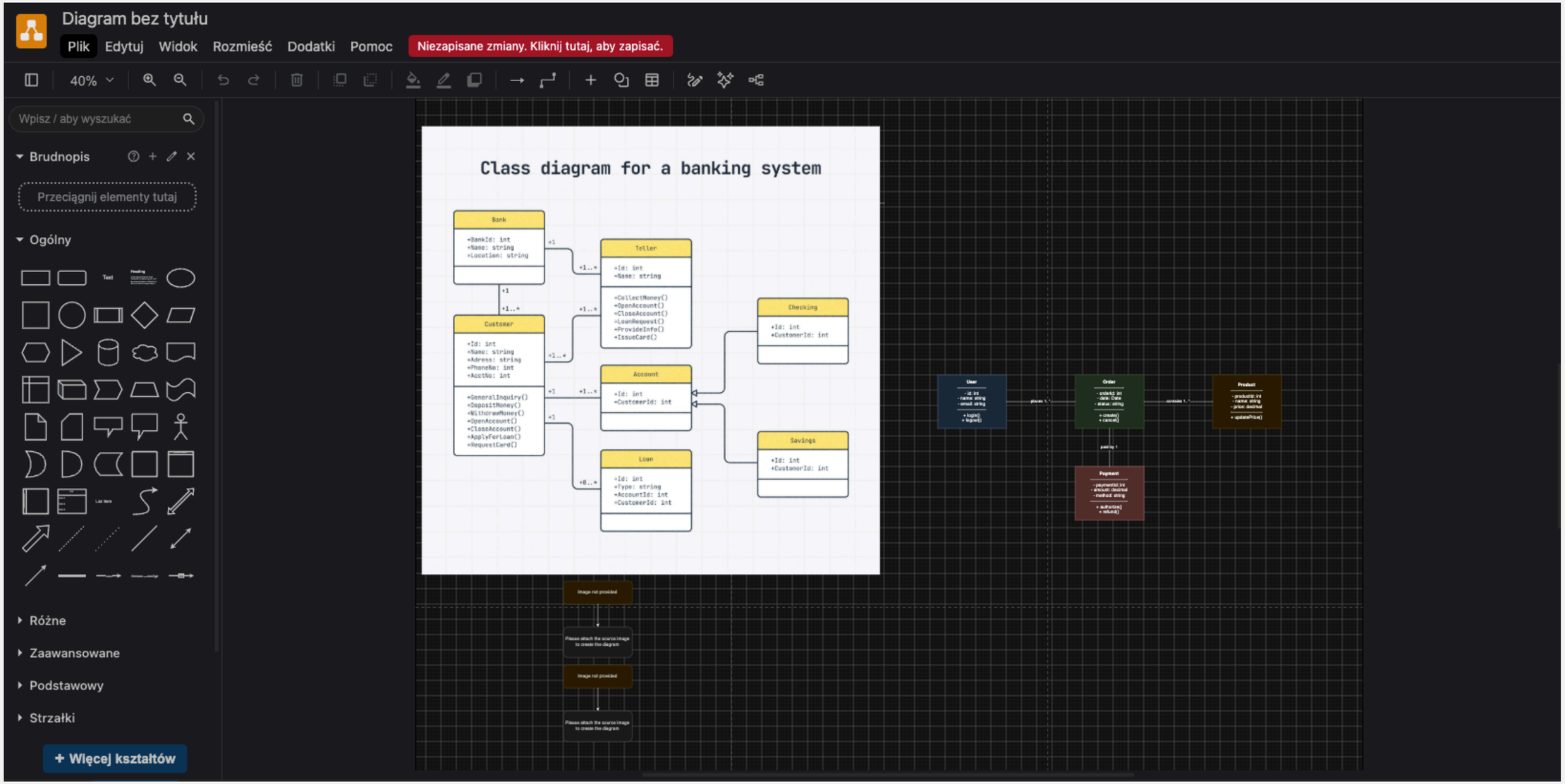
UML Use Case Diagrams

- diagrams.net / draw.io
- Visual Paradigm Online
- PlantUML
- Lucidchart
- Miro / Creately

draw.io for beginners, Visual Paradigm for proper UML, PlantUML for code-oriented engineers.

You may use another UML-capable tool, but your diagram must still meet the task requirements and be included in your project evidence.
And personally I’d recommend:

https://app.diagrams.net



M

M03 → S05 + S06

Task 2 → choose one use case from your diagram
Task 3 → describe its normal flow + alternative/failure flow
Task 4 → model the same flow as an activity diagram

Overview → detailed description → visual workflow

Before you leave — M03 checklist

Required evidence for Tasks 2–4

- ✓ M03_usecase saved
- ✓ M03_activity saved
- ✓ Actor choices briefly explained
- ✓ Decision guards labelled
- ✓ One textual use case saved
- ✓ Activity paths match textual use case
- ✓ Alternative / failure flow included
- ✓ Complete project ZIP exported
- ✓ Requirement ID linked
- ✓ Backup stored outside university computer

No formal Moodle checkpoint today — export your progress ZIP anyway.

THANK YOU